

Sturm - Liouville DE.

$$\frac{d}{dx} \left[p \frac{dy}{dx} \right] - qy + \lambda r y = 0 \quad p(x), q(x), r(x)$$

eigen value
weighting function.

일반적인 2nd order ODE는

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0.$$

위와 같은 형태로 쓰면,

$$\frac{d}{dx} \left[m \frac{dy}{dx} \right] = \frac{dm}{dx} \frac{dy}{dx} + m \frac{d^2 y}{dx^2}$$

$$\frac{d}{dx} \left[m \frac{dy}{dx} \right] - qy + \lambda r y = 0.$$

$$m \frac{d^2 y}{dx^2} + \left(\frac{dm}{dx} \right) \frac{dy}{dx} - qy + \lambda r y = 0.$$

|| 위 아래 곱함.

$$m \frac{d^2 y}{dx^2} + m P \frac{dy}{dx} + m Q y = 0.$$

m, q, r, λ와 P, Q 사이 관계를 찾으면.

$$mP = \frac{dm}{dx} \quad \left| \int P dx = \int \frac{1}{m} dm \right. \\ \left. = \ln m. \right.$$

$$m(x) = e^{\int P(x) dx}.$$

m은 P(x)로 인한 곱셈.
정확히 인자라고 볼 것.

$$mQ = -q + \lambda r.$$

$$Q = \frac{\lambda r - q}{m(x)}$$

연산자 L 정의하자.

$$L(y) = \frac{d}{dx} \left[m \frac{dy}{dx} \right] - qy + \lambda r y = 0$$

$$D(y) = -\frac{d}{dx} \left[m \frac{dy}{dx} \right] + qy$$

$$L(y) = [-D + \lambda r] y = 0$$

그러면 $D(y) = \lambda r y$. y는 D에 대해 eigenfunction, λ는 eigen value.

D는 자기수반 연산자다.

$$\frac{d^2 y}{dt^2} + \frac{c - (a+b+1)t}{t(1-t)} \frac{dy}{dt} - \frac{ab}{t(1-t)} y = 0.$$

$$m = e^{\int P(x) dt}.$$

$$\int P(x) dx = \int \frac{c}{t(1-t)} dx - \int \frac{a+b+1}{1-t} dx.$$

$$= c \ln t - c \ln(1-t) + (a+b+1) \ln(1-t).$$

$$m(t) = t^c (1-t)^{-c} (1-t)^{a+b+1}.$$

$$= \frac{t^c (1-t)^{a+b+1}}{(1-t)^c} = t^c (1-t)^{a+b-c+1}.$$

equivalent equation.

이거? 아니지?

이거 맞지?

$$\frac{d}{dt} \left[t^c (1-t)^{a+b-c+1} \frac{dy}{dt} \right] - t^c (1-t)^{a+b-c+1} \cdot ab t^{-1} (1-t)^{-1} y = 0$$

$$\frac{d}{dt} \left[t^c (1-t)^{a+b-c+1} \frac{dy}{dt} \right] - ab t^{c-1} (1-t)^{a+b-c} y = 0.$$

Sturm-Liouville differential equation.

$$\frac{d}{dt} (t^c (1-t)^{a+b-c+1}) = c t^{c-1} (1-t)^{a+b-c+1} - (a+b-c+1) t^c (1-t)^{a+b-c}$$

$$= (c - c t - a t - b t + c t - t) t^{c-1} (1-t)^{a+b-c} = \{c - (a+b+1)t\} t^{c-1} (1-t)^{a+b-c}$$

$$t^c (1-t)^{a+b-c+1} \frac{d^2 y}{dt^2} + \{c - (a+b+1)t\} t^{c-1} (1-t)^{a+b-c} \frac{dy}{dt} - ab t^{c-1} (1-t)^{a+b-c} y = 0$$

$$t(1-t) \frac{d^2 y}{dt^2} + \{c - (a+b+1)t\} \frac{dy}{dt} - aby = 0. \quad \mathcal{L}[y, \dot{y}]$$

$$\frac{d}{dx} [m(x)]$$

$$\frac{d}{dt} \left[m(t) \frac{dy}{dt} \right] - q(t) y = 0$$

이거 라그랑지안으로 나타내기.

$$\frac{d}{dt} \left[\frac{\partial \mathcal{L}}{\partial \dot{y}} \right] - \frac{\partial \mathcal{L}}{\partial y} = 0.$$

$$\frac{d}{dt} \left[m(t) \dot{y} \right] - q(t) y = 0.$$

$$\downarrow \frac{\partial \mathcal{L}}{\partial \dot{y}} \quad \downarrow \frac{\partial \mathcal{L}}{\partial y}$$

$$\mathcal{L} = \frac{q(t)}{2} y^2 + m(t) \frac{1}{2} \dot{y}^2.$$

$$m(t) = t^c (1-t)^{a+b-c+1}.$$

$$q(t) = ab t^{c-1} (1-t)^{a+b-c}$$

$$\Rightarrow \mathcal{L}(y, \dot{y}, t) = \frac{1}{2} t^c (1-t)^{a+b-c+1} \dot{y}^2 + \frac{1}{2} ab t^{c-1} (1-t)^{a+b-c} y^2.$$

이거 맞지?